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The Role of Overstocking and Poor Stock Rotation in Food Waste in Saudi Supermarkets

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Abstract

This study investigated how overstocking and lack of effective stock rotation contribute to food wastage in Saudi Arabian supermarkets. Although food wastage can be considered a behavioral problem for consumers, the study examined internal business operation practices within supermarkets, including inventory management processes. Specifically, a quantitative cross-sectional survey will be carried out based on a structured questionnaire. The survey targets supermarket workers responsible for managing inventory levels and implementing stock rotation techniques. A total of 13 completed questionnaires were obtained. Descriptive statistics and correlation analyses were performed to analyze the collected data. From the results obtained, it is clear that the issue of overstocking in supermarkets still persists due to overordering and mismatch between supply and demand. However, stock rotation measures such as FIFO are commonly used by most supermarkets, but they may not always be effectively implemented. In this case, overstocking has a more direct effect on the food wastage process compared to stock rotation, which has an indirect effect through increasing product expiration. Food wastage is identified as a moderate problem caused by the internal business environment.

1. Introduction

One of the most acute problems in contemporary food systems remains food wastage, which results in various negative impacts on both economy and society. Although there have been numerous advancements in terms of food production, transportation, and even storage, there are still a lot of edible products that never end up being consumed. Even though wastage takes place on several different stages of the chain, much of it happens at the retailing level, when goods are stored, exhibited, and processed by retailers prior to selling. In this way, food at this particular stage already bears the burden of all investments that have gone into producing and delivering it to its current location.

The most impactful aspect of operations with regards to food waste in retail environments is inventory management. Stores need to ensure that their shelves are always full; however, at the same time, their aim is to avoid spoilage of goods. The issue is particularly challenging in the case of perishable goods including dairy products, fruits and vegetables, as well as baked goods. There are two aspects of operations that should be noted in connection with this matter. First, overstocking involves ordering too much product and therefore having goods which would expire prior to them being sold by the retailer. Second, poor stock

rotation may mean that the oldest products do not get sold because of a failure to use the FIFO approach properly.

In cases where the inventory policy does not align with the demand trends, there is bound to be an oversupply of products, hence a high likelihood of spoilage. The same problem arises when the process of rotating the stock is not properly adhered to, as the result could be that newer items find their way to the front of the store while the older ones remain at the back, thus increasing the possibility of their spoilage. Studies have consistently found that misforecasting, overstocking, and poor shelf-life management are key factors leading to food waste in retail logistics. Conversely, better information systems, proper planning, and stricter rotation policies may help mitigate food waste.

Such problems are particularly significant in countries where there is heavy dependency on imported foodstuffs and difficult environmental conditions prevail. An illustrative case in point would be Saudi Arabia. Due to the high level of dependency on imported foodstuffs, wastage occurring in local retail networks poses a much higher economic risk. Every item that goes waste means not just money wasted for local retailers but inefficiencies occurring throughout the supply chain process. Moreover, Saudi Arabia has seen rapid growth in its retail industry over the past few decades, with many large supermarket networks being responsible for the delivery of food items to the country's urban centers. Such growth in retail networks has made accurate forecasting even more crucial in reducing waste.

Despite the recognition of the problem of food waste in Saudi Arabia by scholars and policymakers, most of the studies conducted have largely focused on issues such as consumption habits, cultural practices, and awareness strategies. Even though these aspects are very crucial, they fail to account for factors associated with logistics operations. The occurrence of food waste at supermarkets results from pre-existing activities such as order volumes, safety stocks, reorder points, and product expiration periods. Numerous studies from around the world reveal that the leading sources of food waste in supermarkets are over-ordering and improper stock rotation. Nevertheless, few studies have empirically measured the contribution of overstocking and poor stock rotation to food waste at Saudi Arabian supermarkets. This is due to differences in supply chains, technology adoption, and climatic conditions from one country to another.

From the perspective of supply chain management, this difference has substantial importance. As retailers form an intermediary position between suppliers and consumers, their decisions about inventory impact production levels, distribution logistics, and how products are handled along the entire chain. In case of households being responsible for more wastage, inefficient management at the retailer level could result in increased wastage upstream and downstream of the chain. Thus, solving the problem of overstocking and inefficient stock rotation will yield benefits for the retailer beyond its immediate business premises.

The main objective of this study is to evaluate the role that overstocking and improper stock rotation play in causing food waste in Saudi retailers. Instead of considering food waste as a wide sustainability issue, this

study attempts to explore the particular actions made by managers, i.e., ordering and stock rotation, that can lead to wastage.

Therefore, the research questions that guide this study include:

To what extent do overstocking and poor stock rotation practices contribute to food waste in Saudi supermarkets?

The importance of this research is twofold. First, from an academic standpoint, the research makes a valuable contribution to the existing yet small literature base that focuses on understanding the relationships between the inventory management practices of retailers and the generation of food waste in the particular country under study and other countries that import most of their products. Second, the results of this research can be useful in helping managers of Saudi supermarkets reduce waste by improving their internal practices rather than focusing on consumers only.

2. Literature Review

2.1 Introduction

Food waste represents one of the most persistent inefficiencies within contemporary food systems, generating economic losses, environmental degradation, and unnecessary pressure on natural resources. Globally, approximately one-third of food produced for human consumption is lost or wasted across supply chain stages (Gustavsson et al., 2011). While public debate often attributes food waste primarily to consumer behaviour, academic research increasingly demonstrates that substantial volumes are embedded within upstream operational decisions, particularly at the retail stage. Supermarkets function as a critical interface between suppliers and consumers, and their inventory management practices significantly influence whether products are sold within their shelf life or discarded.

This literature review critically analyses existing research linking inventory management practices to retail food waste, with specific focus on Saudi supermarkets. It argues that overstocking and weak stock rotation are not isolated managerial errors but predictable outcomes of demand uncertainty, fragmented supply chain coordination, service-level pressures, and limited forecasting integration. From an inventory management perspective, supermarkets operate within a structural trade-off between service level optimisation and holding cost minimisation; however, in perishable systems this trade-off extends beyond cost into waste generation, revealing tension between traditional buffering logic and sustainability objectives. The review is organised around three themes: retail food waste as an operational issue, forecasting and overstocking dynamics, and stock rotation practices within the Saudi retail context.

2.2 Retail Food Waste as an Operational Issue

Early global assessments quantify the magnitude of food waste but provide limited explanation of operational drivers. Gustavsson et al. (2011) demonstrate that retail contributes significantly to waste volumes in higher-income systems. However, macro-level aggregation does not identify the specific store-level decisions that translate surplus into waste. More recent research moves toward operational explanations. Todd and Faour-Klingbeil (2024) argue that handling practices, shelf-life mismanagement, and inventory errors substantially influence discard rates at retail and foodservice levels.

Logistics-oriented studies further support the operational perspective. Nikoličić et al. (2021) demonstrate that inefficiencies in replenishment coordination and logistics timing increase spoilage risk in retail supply chains. Similarly, Aljohani (2023) shows that optimising distribution networks in perishable systems reduces food waste through improved routing efficiency and reduced transportation time. These studies confirm that operational design affects waste outcomes; however, they primarily focus on upstream logistics rather than in-store inventory discipline.

In the Saudi context, Mir et al. (2024) identify environmental, economic, and infrastructural contributors to food waste but acknowledge limited empirical examination of supermarket-level practices. Oelofse et al. (2020) provide comparative analysis across regions, yet focus largely on macro-system drivers rather than store-specific decisions. As a result, although scholars broadly agree that retail contributes significantly to food waste, debate persists regarding whether waste originates primarily from systemic supply chain constraints or from internal inventory management decisions. The literature increasingly supports the view that retail waste is operationally produced, yet empirical isolation of specific practices such as overstocking thresholds and FIFO compliance rates remains limited.

2.3 Demand Uncertainty, Forecasting, and Overstocking

Two operational practices are especially relevant in this context: overstocking and poor stock rotation. Overstocking refers to maintaining inventory quantities beyond realistic demand requirements, typically as a safeguard against stockouts and lost sales. Poor stock rotation, particularly weak adherence to first-in, first-out (FIFO) procedures, results in preventable product expiration even when customer demand exists. In import-dependent countries such as Saudi Arabia, where food products travel long distances and climatic conditions shorten effective shelf life, these inefficiencies carry amplified financial and environmental consequences.

Inventory management theory conceptualises stock as a buffer against demand variability. Maintaining higher safety stock levels is frequently justified as a rational response to forecast error and demand volatility. However, in perishable systems, this buffering strategy can generate unintended waste consequences. Alsoussi and Tahboub (2025) find that organisations often rely on informal judgment rather than advanced analytical forecasting tools, leading to inflated safety stock levels. Their findings suggest that precautionary over-ordering emerges not only from uncertainty but also from limited analytical capability.

Within Saudi Arabia, Abunar (2012) identifies fragmented coordination between supermarkets and suppliers, with weak information sharing increasing demand uncertainty. In such environments, conservative ordering behaviour becomes a risk-management strategy. Abunar and Zerban (2016) argue that improved accounting information systems enhance supply chain coordination and visibility, yet the relationship between improved coordination and measurable waste reduction remains indirect.

Recent research emphasises technological intervention. Khan et al. (2025) demonstrate that advanced information systems mediate demand uncertainty and significantly improve inventory performance in supermarket contexts. Petrisia and Kularatne (2024) show that machine learning tools reduce perishable waste by improving forecast accuracy and responsiveness to demand fluctuations. Vukolić et al. (2025) similarly argue that artificial intelligence enhances food management efficiency and reduces excess inventory.

Despite these advances, a methodological tension remains. Supply chain studies frequently prioritise turnover rates, service levels, and cost efficiency without directly measuring waste volumes. Conversely, food waste research quantifies discard levels but rarely analyses the forecasting or safety stock decisions that generate surplus. Ahmed (2021) demonstrates that supply chain management practices improve customer satisfaction in Saudi retail firms, yet sustainability metrics remain secondary. Consequently, although forecasting innovation is widely presented as a solution to overstocking, empirical evidence directly linking forecast error margins and safety stock policies to documented waste volumes in Saudi supermarkets remains limited.

2.4 Stock Rotation and Operational Compliance

While overstocking concerns ordering quantities, stock rotation concerns execution within stores. FIFO principles are widely recognised as fundamental to perishable inventory management; however, implementation consistency varies across retail environments. Alsoussi and Tahboub (2025) note that limited technological monitoring, staff training gaps, and time pressures reduce adherence to formal inventory control systems. In high-volume supermarket settings, restocking speed may be prioritised over careful repositioning of older items, increasing expiration risk.

Bhatti and Iqbal (2024) find that disciplined inventory management practices significantly reduce food waste in foodservice settings, suggesting that structured rotation procedures can produce measurable outcomes. In Saudi Arabia, Al-Khateeb et al. (2021) highlight national efforts to address food waste but acknowledge operational challenges in implementation. Alshabanat et al. (2021) quantify national food loss and waste but do not isolate FIFO compliance as a specific driver. Baig et al. (2019) identify behavioural and structural contributors to food waste and call for stronger organisational-level research.

Methodologically, much of the literature relies on qualitative case studies, survey-based assessments, and macro-level estimations. Quantitative store-level audits linking FIFO compliance rates to documented

disposal volumes remain scarce. Although Maqbalia (2024) demonstrates that logistics management affects shelf life in distribution systems, empirical investigation of in-store rotation discipline remains limited. Thus, stock rotation is widely acknowledged as important, yet its measurable contribution to waste generation within Saudi supermarkets remains underexplored.

2.5 Structural Vulnerability in Import-Dependent Systems

Saudi Arabia's import-dependent food system intensifies the consequences of inventory inefficiencies. The economic cost of discarding imported food extends beyond retail margins, reflecting wasted transportation, storage, and energy inputs embedded throughout the supply chain. Extended international supply chains reduce remaining shelf life upon arrival at retail outlets (Abunar, 2012). Climatic conditions further accelerate spoilage risk, increasing reliance on accurate forecasting and disciplined rotation. Mir et al. (2024) highlight environmental and economic pressures associated with food waste in the Kingdom, emphasising the amplified cost of inefficiency in resource-constrained environments.

While data-driven optimisation models have been tested in European and Australasian retail systems (Petrisia & Kularatne, 2024), comparable empirical investigation within Saudi supermarkets remains limited. Mmereki et al. (2024) argue that effective waste prevention in middle-income contexts requires operational reform rather than solely behavioural change, reinforcing the need to examine supermarket-level decision-making.

Despite contextual urgency, store-level quantitative research isolating the contribution of overstocking and FIFO non-compliance within Saudi supermarkets remains limited. The literature recognises operational inefficiencies but lacks precise empirical measurement linking inventory policies to documented waste outcomes.

2.6 Synthesis and Gap Identification

Across the literature, several areas of agreement emerge. Retail food waste is structurally embedded in supply chain and inventory practices rather than solely consumer behaviour. Demand uncertainty and fragmented coordination encourage precautionary overstocking, while weak rotation discipline increases expiration risk. Technological integration and forecasting innovation are widely promoted as corrective mechanisms.

However, significant gaps persist. Forecasting studies frequently rely on efficiency indicators rather than direct waste measurement. FIFO compliance is discussed normatively but rarely assessed through systematic quantitative audit. Saudi research remains predominantly macro-level, with limited store-specific empirical investigation. Moreover, few studies integrate forecasting analysis, safety stock policy evaluation, and rotation compliance within a single empirical framework.

Therefore, although theoretical connections between overstocking, poor stock rotation, and food waste are well articulated, empirical isolation of their measurable impact within Saudi supermarkets remains insufficient. Addressing this gap is necessary to move from conceptual recognition toward evidence-based operational reform.

2.7 Conclusion

The reviewed literature demonstrates that retail food waste is embedded within inventory management systems rather than solely consumer behaviour. Overstocking reflects rational buffering under uncertainty yet increases expiration risk in perishable environments. Weak stock rotation transforms surplus inventory into preventable waste. In Saudi Arabia's import-dependent context, these inefficiencies carry heightened economic and environmental consequences.

Without isolating the operational contribution of overstocking and FIFO non-compliance, policy and technological interventions risk addressing symptoms rather than structural drivers of waste. Direct store-level measurement linking ordering practices, safety stock thresholds, rotation compliance, and documented waste data remains limited.

Future research should adopt quantitative operational designs capable of measuring the causal contribution of overstocking and poor stock rotation within Saudi supermarkets. Such evidence is necessary to inform managerial reform, enhance supply chain efficiency, and support the development of more sustainable retail food systems in the Kingdom.

3. Methodology

3.1 Introduction

This chapter describes the research design used to examine the effect of overstocking and poor stock rotation on food waste in Saudi supermarkets. The study adopted a **quantitative cross-sectional survey design**, using numerical data collected from supermarket employees through a structured questionnaire (Creswell, 2014; Saunders et al., 2019).

3.2 Research Design

A quantitative, cross-sectional survey design will be adopted in this study, where the correlation between overstocking, stock rotation, and food waste is to be established. Quantitative research enables statistical analysis to be performed on structured data, which helps establish trends and correlations (Creswell, 2014; Bryman, 2016).

The data collection was done at one point in time, providing insight into existing processes within the organization (Saunders et al., 2019). The survey design helped obtain responses from workers with various roles in the company and can be considered an appropriate choice in this case (De Vaus, 2013).

3.3 Participants and Sample

Target population consists of supermarket employees working in Saudi Arabia in terms of inventory and stock management, including managerial, supervisory, and stocking staff members.

Sampling techniques used in this research include non-probability sampling, purposive, and convenience sampling (Etikan et al., 2016; Taherdoost, 2016).

Participants were expected to satisfy the following criteria:

1. Work in a supermarket or retail food setting within Saudi Arabia
2. Have responsibilities related to inventory or stock management

The study aimed to collect 50–100 responses; however, only 13 were obtained due to time and access constraints.

3.4 Data Collection Method and Strategy

Data collection was done using an instrument that measured overstocking, stock rotation, and food waste.

The questionnaire consisted of demographic questions and employed the five-point Likert scale for uniformity (Likert, 1932). It was sent out using Google Forms, and it was sent out to LinkedIn and WhatsApp contacts, including supermarket staff members. Survey participation was optional and anonymous, with the results being utilized for academic purposes only. Surveys are effective in gathering data and uncovering patterns within individuals (De Vaus, 2013).

3.5 Data Analysis

Analysis of data involved the use of descriptive statistics such as frequency, mean, and percentage (Field, 2013).

Correlation analysis was applied to determine the existence of relationships between overstocking, stock rotation, and food waste. With the limited sample size, analysis did not delve into advanced relationships but only simple statistics (Hair et al., 2019).

3.6 Limitations

However, this research study is prone to some limitations. Firstly, there are chances that employing a non-probability sampling strategy might affect the representativeness of data collected and make the study vulnerable to selection bias (Taherdoost, 2016). Secondly, the number of individuals who participated in the study was very small, totaling 13 participants. This has weakened the ability of findings from this study to be

generalized into the wider supermarket industry in Saudi Arabia. In addition, this limitation weakens the power of statistical tests used for data analysis and makes each individual respondent more significant than is desired (Bryman, 2016). Lastly, since this study used cross-sectional survey methodology, there is no possibility of establishing cause-and-effect relationship among the variables investigated (Creswell, 2014).

RESULTS AND ANALYSIS

4.1 Introduction

This chapter highlights the research findings from the survey data analysis regarding the phenomenon of overstocking, stock rotation, and food wastage among supermarkets. These findings will be discussed under demographics, independent variables, and dependent variables. It is noted that due to the smaller sample size, findings should be carefully considered.

4.2 Demographic Profile of Respondents

There were a total of 13 replies received.

It is observed that the participants belonged to the operations and supervisory levels and were therefore competent in handling inventory operations.

Most of the participants were working in grocery (38.5%) and bakery sections (30.8%). Others were in fresh, dairy, and meat sections. More than half of the participants had between one and three years of experience (61.5%).

Most respondents were supervisors (46.2%), followed by store managers (23.1%), while inventory staff and department heads each accounted for 15.4%.

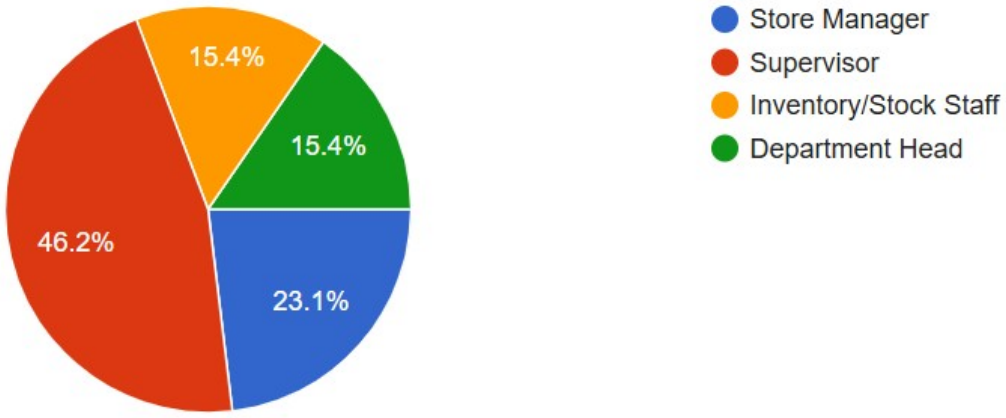


Figure 4.1: Distribution of Job Roles

Figure 4.1 shows that the majority of respondents are supervisors, indicating strong representation of operational-level employees.

Respondents were mainly from grocery (38.5%) and bakery (30.8%), with smaller representation from fresh produce, dairy, and meat.

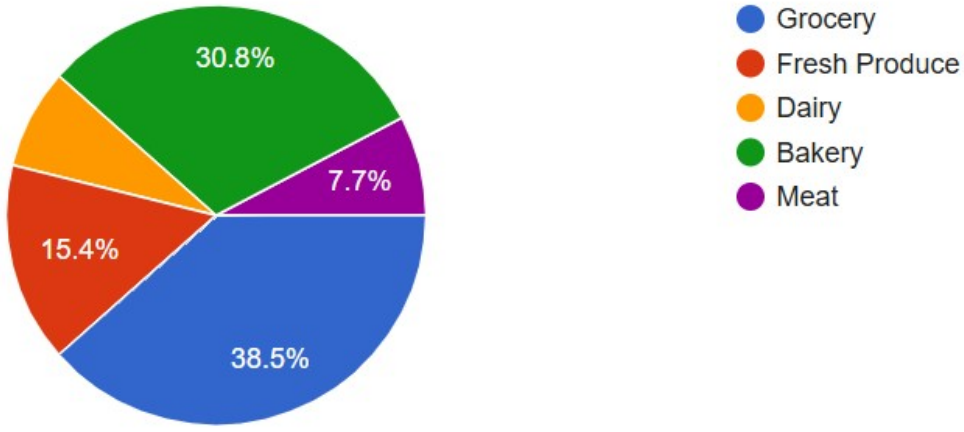


Figure 4.2: Distribution of Departments

Figure 4.2 shows that most respondents work in grocery and bakery sections, reflecting areas with high involvement in perishable inventory management.

Most respondents had between 1–3 years of experience (61.5%), indicating practical, hands-on involvement in supermarket operations.

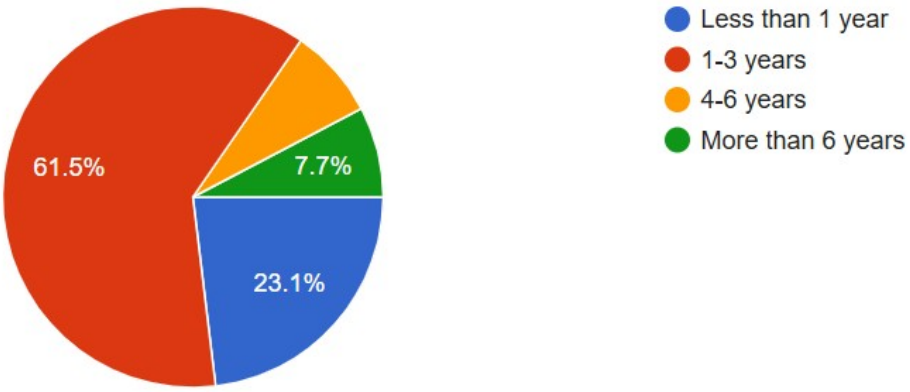


Figure 4.3: Years of Experience of Respondents

Figure 4.3 shows that the majority of respondents have moderate experience, supporting the reliability of their responses.

4.3 Overstocking Practices

Most participants agreed that stock levels are generally aligned with customer demand. Nevertheless, most of them admitted that there was stock surplus and overordering, implying that there were discrepancies in the planning and ordering processes.
This means that even if the inventory management system exists, it is not fully utilized during implementation. Therefore, surplus stock can be amassed, creating opportunities for product loss. Ultimately, overstocking is a recurring problem associated with inefficient operations.

Many respondents indicated that overstocking occurs within their departments, suggesting inconsistencies between planning and actual ordering.

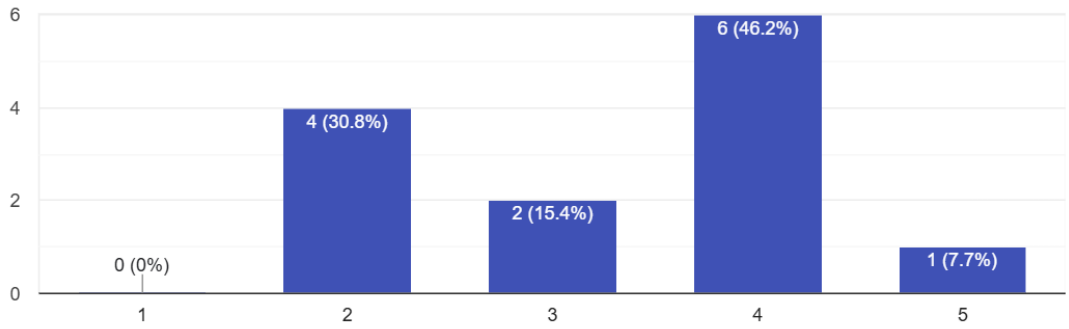


Figure 4.4: Perception of Overstocking in Stores

Figure 4.4 shows that overstocking is a recurring issue, reflecting inefficiencies in inventory planning and demand forecasting.

4.4 Stock Rotation Practices

The findings indicate that inventory management processes like FIFO and checking expiry dates are adhered to. This implies that there is knowledge among the employees regarding inventory management procedures. Inconsistency in answers received implies that not all branches are doing it the same way. This might be due to variations in training, supervision, and control among others. Hence, the problem does not lie in the system but in its implementation.

Stock rotation practices such as FIFO are generally followed, although some variation in responses indicates inconsistency in implementation.

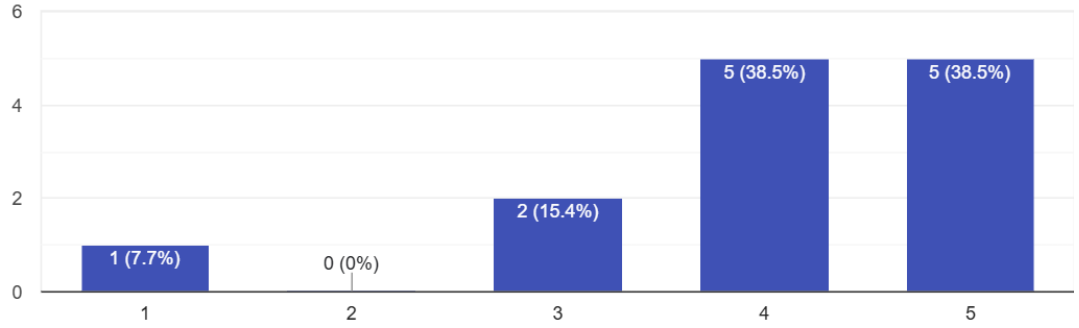


Figure 4.5: Implementation of FIFO Practices

Figure 4.5 shows that while FIFO is commonly practiced, it is not consistently applied across all stores.

4.5 Food Waste

It is evident that food waste is considered to be an issue of medium level that persists, as the answers range from neutral to agree. In other words, it appears that despite not being severe, food waste is an existing operational problem. The conclusion is that food waste is a function of internal practices.

Food waste was reported as a moderate but recurring issue in supermarket operations.

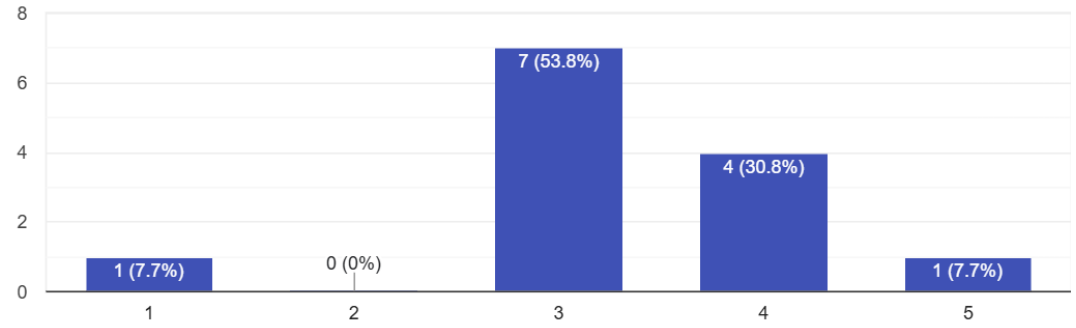


Figure 4.6: Frequency of Food Waste in Stores

Figure 4.6 shows that food waste is a consistent operational challenge, influenced by internal practices.

4.6 Relationship Between Overstocking, Stock Rotation, and Food Waste

The results provide a direct link between the process of inventory management and food waste.

Increased instances of overstocking and overordering have been directly linked with food waste, and this is an indication that surplus inventory causes food wastage due to expired foods. In this case, it is evident that ineffective orders cause more food wastage than other factors.

While the practice of stock rotation is normally practiced by most supermarkets, inconsistency in stock rotation often leads to expiry of stock. While this is the case, overstocking seems to have a more powerful impact than improper stock rotation.

4.7 Additional Insights from Open-Ended Responses

The open-ended questions gave additional information about the reasons for food wastage, such as:

- unfavorable storage conditions
- uncertainty about the expiration date
- failure to communicate effectively
- treatment of perishables

This reveals that food wastage does not occur because of one reason alone but is the result of multiple operational problems in supermarkets.

4.8 Summary of Findings

The results show that both overstocking and stock rotation practices influence food waste.
While supermarkets attempt to match supply with demand, over-ordering and excess stock remain common. Stock rotation practices are generally in place but are not always consistently applied. Together, these factors contribute to food waste, which remains a moderate but persistent issue in supermarket operations.

DISCUSSION AND CONCLUSION

5.1 Introduction

This chapter discusses the study's findings in relation to the research question and literature. It also presents practical and theoretical implications, limitations, and recommendations for future research.

5.2 Discussion of Findings

The results indicate that both overstocking and stock rotation are causes of food waste, although with varying effects.
It is noted that overstocking was common, with goods usually being ordered more than necessary. This reinforces other studies by Teller et al. (2018) and Eriksson (2015), which attribute improper forecasting and replenishment to waste generation.
While FIFO was generally practiced, some inconsistencies were observed in how it was done. This agrees with the work of Mattsson & Williams (2022), who argue for the contribution of employee practice to waste.
It emerged that waste is significant, though moderate, which suggests that inefficiency exists despite management measures in place. Open-ended questions reveal problems like inefficient storage, incorrect expiry date interpretation, and miscommunication.
In all, overstocking plays a more direct role than poor stock rotation.

5.3 Practical Implications

Demand forecasts and orders must be improved in supermarkets to ensure there is no surplus. The importance of proper stock turnover procedures cannot be underestimated, especially by means of increased training and supervision.
Effective communication among employees will decrease mistakes, and an improved system for managing stocks will make decision-making easier.

5.4 Theoretical Implications

The current work adds to the literature on OSCM as it places the importance of retail store-level activities on food waste issues. The current literature tends to pay more attention to consumers' behaviors; however, the above results show how crucial inventory and human activities can be.
Moreover, it indicates that simply developing a good system is not sufficient – its implementation is crucial too.

5.5 Limitations of the Study

There are various limitations in this study. Firstly, there is a low number of respondents (13), and non-probability sampling technique is used, making this study difficult to generalize. Self-reporting may introduce biased results. Also, cross-sectional analysis cannot be used to determine causation.
Another limitation is that the study is based on employees' perception and not factual data.

5.6 Recommendations for Future Research

Future studies should use larger and more diverse samples and consider mixed methods to gain deeper insights. Research could also focus on specific departments, such as fresh produce, where waste is more critical.
Further studies should explore the role of technology and real operational data in improving inventory management and reducing food waste.

5.7 Conclusion

This study examined how overstocking and stock rotation contribute to food waste in Saudi supermarkets. The findings show that while stock rotation practices exist, inconsistent implementation and overstocking lead to waste. Overstocking has a stronger impact, while poor rotation plays a supporting role.
Overall, improving inventory management, operational discipline, and staff coordination is essential to reducing food waste in supermarket operations.

4. Findings (Results)

4.1 Introduction

This chapter presents the key findings obtained from the questionnaire distributed to supermarket employees in Saudi Arabia. A total of 13 valid responses were collected and analysed. The results have been categorized under five headings: demographics of respondents, over-stocking, stock turn-over, food waste, and other operational issues.

4.2 Demographics of Respondents

One clear trend that emerged from the analysis was the presence of a large number of supervisory and managerial personnel. As a matter of fact, supervisors made up the highest percentage at 46.2%, while store managers were second at 23.1%. Inventory or stock staff and department heads each accounted for 15.4%. Across the dataset, respondents were mainly individuals involved in store operations and inventory-related responsibilities.

The second trend identified from the findings was that of employees located in grocery and bakery sections. This is clearly shown from the 38.5% in grocery and 30.8% in bakery. Those from fresh produce accounted for 15.4%, and those from dairy and meat accounted for 7.7%. Across the dataset, responses were received from departments that regularly handle food inventory.

Another pattern observed in the data was the moderate level of work experience among participants. This can be supported with 61.5% of respondents indicating experience from one to three years. Furthermore, 23.1% of the respondents indicated experience of less than one year, while 7.7% indicated experience of four to six years, as well as another 7.7% indicated experience greater than six years.

4.3 Overstocking Practices

A significant pattern observed in the data was generally positive responses regarding stock levels matching customer demand. This is proven by the fact that 46.2% of those who responded chose 'agree' and 7.7% chose 'strongly agree.' On the other hand, 30.8% chose 'disagree,' while 15.4% chose 'neutral.' The data set had answers that spanned. Another pattern observed in the data was mixed responses regarding whether overstocking was common in respondents' departments. This is evidenced by 23.1% selecting strongly disagree, 23.1% selecting disagree, 30.8% selecting neutral, and 23.1% selecting agree. No respondents selected strongly agree. Across the dataset, views on overstocking varied between departments.

A further pattern observed in the data was varied responses regarding products being ordered in quantities exceeding customer demand. This is evidenced by 15.4% selecting strongly disagree, 30.8% selecting disagree, 15.4% selecting neutral, and 38.5% selecting agree. No respondents selected strongly agree. Across the dataset, responses showed both agreement and disagreement.

Another pattern observed in the data was positive responses toward inventory monitoring and adjustment practices. This is evidenced by 38.5% selecting agree and 15.4% selecting strongly agree. In addition, 38.5% selected neutral, while 7.7% selected strongly disagree. Across the dataset, most respondents reported the presence of inventory monitoring activities.

4.4 Stock Rotation Practices

A significant pattern observed in the data was strong agreement regarding FIFO implementation. This is evidenced by 38.5% of respondents selecting agree and 38.5% selecting strongly agree. In addition, 15.4% selected neutral, while 7.7% selected strongly disagree. Across the dataset, most responses were concentrated in the agreement categories.

Another pattern observed in the data was positive responses regarding prioritising older products over newer products on shelves. This is evidenced by 23.1% selecting agree and 38.5% selecting strongly agree. In addition, 30.8% selected neutral and 7.7% selected disagree. Across the dataset, agreement responses formed the largest group.

A further pattern observed in the data was positive responses regarding checking expiry dates during shelf replenishment. This is evidenced by 38.5% selecting agree and 23.1% selecting strongly agree. In addition, 23.1% selected neutral, while 7.7% selected strongly disagree and 7.7% selected disagree. Across the dataset, responses were mainly concentrated in the positive categories.

4.5 Food Waste

A significant pattern observed in the data was the concentration of responses in the neutral category regarding food waste being common in stores. This is evidenced by 53.8% of respondents selecting neutral. In addition, 30.8% selected agree, 7.7% selected strongly agree, and 7.7% selected strongly disagree. Across the dataset, responses were mainly neutral with some positive responses.

Another pattern observed in the data was mixed responses regarding whether overstocking occurred frequently in departments. This is evidenced by 23.1% selecting strongly disagree, 15.4% selecting disagree, 30.8% selecting neutral, and 30.8% selecting agree. No respondents selected strongly agree. Across the dataset, responses were spread across several categories.

4.6 Coordination and Additional Operational Factors

A significant pattern observed in the data was moderately positive responses regarding coordination between ordering, stocking, and sales activities. This is evidenced by 33.3% selecting agree and 16.7% selecting strongly agree. In addition, 33.3% selected neutral and 16.7% selected disagree. No respondents selected strongly disagree. Across the dataset, responses were concentrated in the neutral and positive categories.

Additional open-ended responses identified several operational issues linked to food waste. These responses included poor storage, confusion about dates, frozen food issues, and miscommunication. Across the dataset, multiple operational concerns were reported.

4.7 Summary of Findings

The findings showed that respondents were mainly supervisors, store managers, and inventory-related staff working across grocery, bakery, and other food departments. Responses indicated the presence of inventory monitoring and stock rotation practices, particularly FIFO and expiry-date checking. Responses regarding overstocking and ordering quantities were mixed across categories. Food waste responses were mainly neutral, with additional operational issues such as storage problems, date confusion, and miscommunication also reported.

5. DISCUSSION

5.1 Introduction

This chapter interprets the findings presented in Chapter 4 and relates them to the research question concerning the role of overstocking and poor stock rotation in food waste within Saudi supermarkets. The analysis is linked to the existing literature and operational/management theories regarding inventory management issues. In addition, implications for practice, study limitations, and directions for further research are mentioned.

5.2 Overstocking as a Contributor to Food Waste

It can be stated that overstocking still persists in the analyzed supermarkets since some respondents said that there may be times when goods are purchased in excess amounts relative to customer needs and that there is overstocking in some departmental areas (see Chapter 4.3). This supports Gustavsson et al. (2011), who identified retail inventory imbalance as an important cause of food waste. It also extends Todd and Faour-Klingbeil's (2024) argument that poor stock management increases losses at the retail stage by showing that staff perceptions within Saudi supermarkets reflect similar operational concerns. While these studies discuss food waste broadly, the present findings highlight over-ordering as a specific store-level mechanism.

From the perspective of Inventory Management Theory, safety stock is maintained to prevent stockouts and preserve service levels. However, when buffering decisions exceed realistic demand requirements, perishable goods face a higher probability of expiry before sale. Therefore, the findings highlight that overstocking can transform a protective inventory strategy into a source of operational waste.

5.3 Stock Rotation and FIFO Discipline

The findings indicate that FIFO practices, shelf prioritisation of older products, and expiry-date checks were generally reported positively by respondents (see Section 4.4). This suggests that many participating stores have formal rotation procedures designed to reduce spoilage and improve stock movement.

This supports Bhatti and Iqbal (2024), who found that disciplined inventory procedures reduce food wastage in operational settings. Similarly, Nikoličić et al. (2021) argued that waste reduction depends heavily on efficient internal retail processes rather than external supply factors alone. However, the present findings also include neutral and disagreement responses, indicating that procedure existence does not necessarily guarantee uniform execution.

Through the lens of the FIFO Inventory Rotation Model, older stock should move first so that shelf life is maximised and expiration risk is minimised. Where FIFO is inconsistently applied, newer stock may replace older stock on shelves, delaying the sale of earlier inventory. Therefore, the findings highlight that stock rotation systems are valuable only when implemented consistently in day-to-day practice.

5.4 Food Waste as a Persistent Operational Issue

The findings indicate that food waste was viewed as a moderate but continuing issue, with many respondents selecting neutral responses and a notable proportion agreeing that food waste was common in their stores (see Section 4.5). This implies that waste could actually be normalized in some supermarket settings as opposed to viewing waste as an extraordinary occurrence.

This is in line with the view presented by Mir et al. (2024) that food waste is a continuous challenge in Saudi Arabia both economically and environmentally. This is consistent with the views presented by Al-Khateeb et al. (2021) that food waste remains prevalent regardless of the growing awareness on the issue. While earlier studies often focus on policy or consumer behaviour, the present findings redirect attention toward internal retail operations.

From a Lean Operations Perspective, any activity that consumes resources without creating customer value is considered waste. Waste resulting from discarded edible products is characterized by waste of purchasing costs, labor, storage, and time. The results indicate that even small recurring food wastage can be considered as operational inefficiency.

5.5 Coordination and Communication Failures

The findings indicate that coordination between ordering, stocking, and sales functions was only moderately positive, while open-ended responses mentioned poor storage, confusion regarding dates, frozen food issues, and miscommunication (see Section 4.6). This suggests that food waste is influenced not only by quantities ordered, but also by the quality of internal coordination.

This supports Abunar (2012), who found that fragmented coordination can weaken food retail supply chain performance in Saudi Arabia. It also extends Abunar and Zerban’s (2016) argument that stronger information

systems improve operational visibility, by showing that frontline employees continue to identify communication gaps as a practical issue.

From the perspective of Supply Chain Coordination Theory, operational performance improves when functions share timely and accurate information. If ordering teams, storage staff, and shelf personnel act independently, inventory decisions become disconnected from real store conditions. Therefore, the findings highlight that communication failures can indirectly generate food waste even when stock policies appear reasonable.

5.6 Implications for Saudi Supermarkets

The findings indicate that reducing food waste requires both inventory control and operational discipline. Overstocking must be addressed through better demand forecasting, while stock rotation procedures should be monitored regularly to ensure consistent execution.

This study is consistent with the findings by Khan et al. (2025) regarding improved inventory management with enhanced information systems when dealing with unpredictable demands. This finding further strengthens the conclusion by Petrisia and Kularatne (2024) about the use of forecasting models in decreasing perishable waste within retail stores.

Incorporating the concept of Data-Driven Decision Making, supermarkets can take into account the sales history, expiration date of products, and demand trends to optimize their ordering process. Hence, it becomes evident that the problem of waste needs to address both employee practice and technological assistance rather than either separately.

5.7 Limitations of the Findings

The findings should be interpreted in light of several limitations. Firstly, a limited sample size was used in the study, involving only 13 participants, thus reducing its generalisability beyond the studied supermarkets. Secondly, the study employed employees' opinions rather than waste reports or inventory records. Thirdly, the survey took place in one instance, without considering seasonal differences in demands.

Despite the above-discussed shortcomings, the study results provide exploratory information regarding the role that supermarket operations play in food waste production.

5.8 Recommendations for Future Research

Future research should involve a bigger sample of respondents in several supermarket chains located in various Saudi cities. Researchers could use both survey methods and measure the amount of waste produced directly through inventory records or other means. Another aspect to consider is comparative research conducted among different supermarket departments, like bakery, dairy, groceries, and fruits & vegetables.

Moreover, future research should focus on studying the effects of forecasting applications, artificial intelligence, and automatic expiry alert mechanisms on food waste minimisation.

5.9 Chapter Summary

In general, these results reveal that overstocking, as well as inadequate stock rotation, lead to food waste in Saudi Arabia. However, their impact differs as overstocking leads to a higher amount of excess inventory. On the other hand, inadequate FIFO results in increased risks of expiry of items. Miscommunication, along with inadequate collaboration among employees, adds to inefficiencies and makes it easier for companies to incur losses.

6. CONCLUSION

This study examined the role of overstocking and poor stock rotation in contributing to food waste within Saudi supermarkets. The research revealed that food waste does not depend only on the behavior of consumers, but it is also affected by internal processes within supermarkets. Despite the fact that many participants indicated the existence of monitoring systems of stocks, application of FIFO system and checking expiration dates, it was found that there are still problems related to overordering, incorrect storage of products, poor stock management and communication issues.

This study answers the research question by showing that both overstocking and poor stock rotation contribute to food waste in Saudi supermarkets, although through different mechanisms. Overstocking increases the volume of surplus inventory that may remain unsold until expiry, particularly in perishable product categories. Poor stock rotation contributes when older items are not moved efficiently, increasing the likelihood of spoilage or disposal. Based on the responses collected, overstocking appeared to have the stronger influence, while weak stock rotation acted as an additional contributing factor.

The study contributes to the existing literature by focusing specifically on store-level operational causes of food waste within the Saudi Arabian supermarket context. Most of the discussions revolving around the topic of food waste tend to focus on consumer behavior, educational programs, or national-level sustainability goals. This study, on the other hand, focuses on the impact of decisions related to inventory, as well as regular procedures and coordination within departments. Thus, the study offers valuable exploratory insights regarding the way common practices used in supermarkets can produce or prevent food waste.

From the results presented above, it is clear that supermarkets could minimize avoidable food waste by better demand forecasting, better control over order sizes, better application of FIFO principles, and improved coordination among ordering staff, storage workers, and employees working at shelves. Nevertheless, the current study suffers from several limitations, which include the relatively small number of participants, as

well as reliance on their self-reports. Further research would benefit from using larger samples in several supermarkets in different cities in Saudi Arabia. Also, further research should make use of waste record data, inventory information, and observations to generate a better understanding of the issue under discussion.

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